

Telecommunications

Practical VoIP Implementation



FreePBX®
let freedom ring™

1. Introduction

Sirs Systems is a small not-for-profit corporation supporting start-up enterprises through remote and virtual working techniques. The organization seeks to minimise operating costs by transitioning towards a regionally distributed workforce model employing Voice over Internet Protocol (VoIP) technologies.

The goal of this practical implementation was to create and configure a prototype VoIP system capable of supporting distant workers utilising FreePBX and Asterisk technologies. The installation aims to provide essential business communication capabilities including SIP extensions, voicemail, Interactive Voice Response (IVR), Ring Groups, conferencing facilities and VoIP security procedures.

The system was implemented utilising a Debian 12 virtual machine running FreePBX within VMware Workstation 17 Pro. Multiple SIP softphones were established across Windows and Android devices to simulate geographically spread consumers operating across remote locations.

2. Network Design

2.1 Overview of Network Architecture

The VoIP system was based around a centralised FreePBX server installed on a Debian 12 virtual machine. SIP softphones operating on Windows and Android devices were linked to the PBX server via the local network environment.

The implemented solution provides:

- Remote SIP extensions
- Voicemail services
- IVR functionality
- Ring Groups
- Audio conferencing
- Blacklisting
- VoIP security configuration
- Secure SIP transport support
- Music on Hold

2.2 Network Diagram

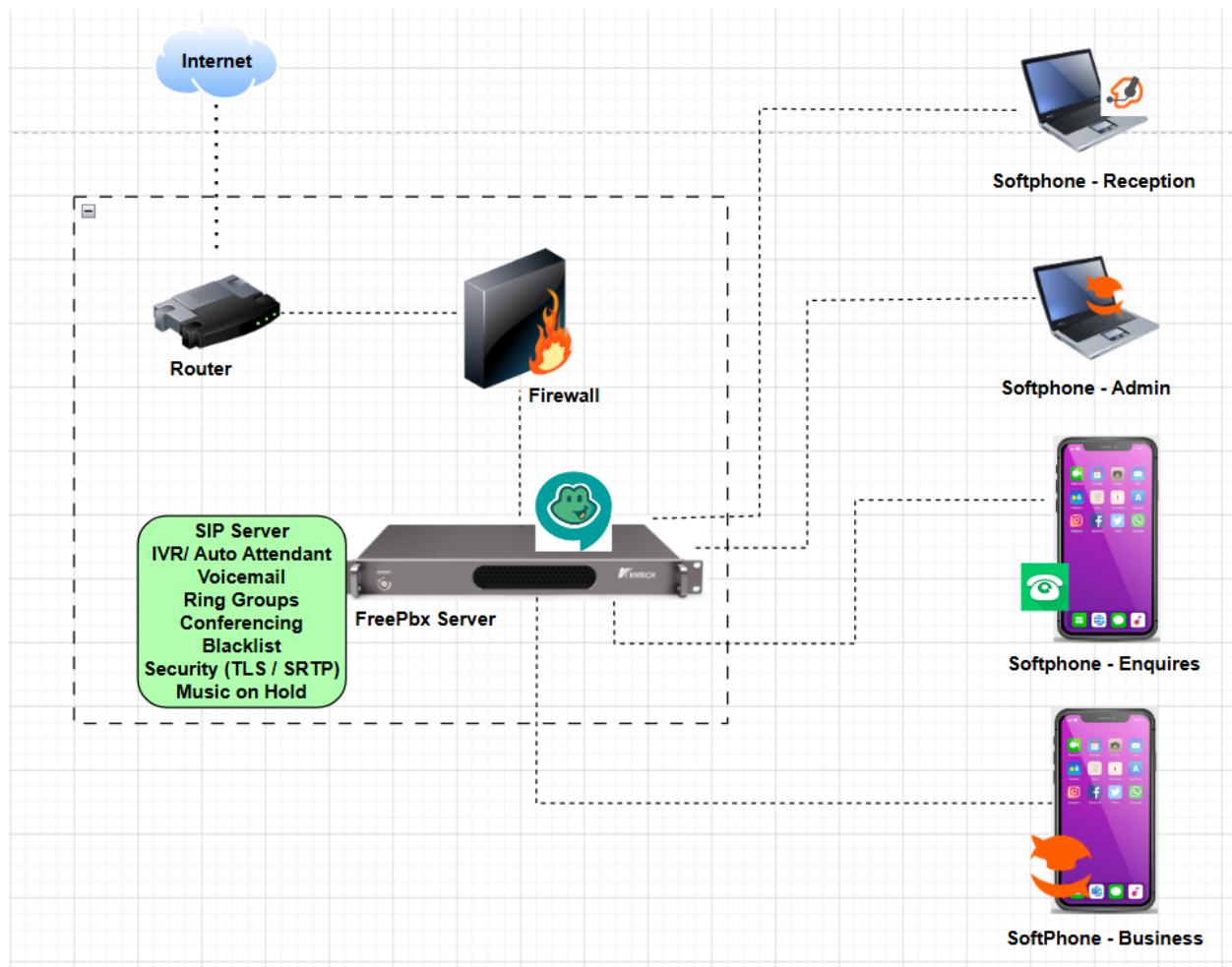
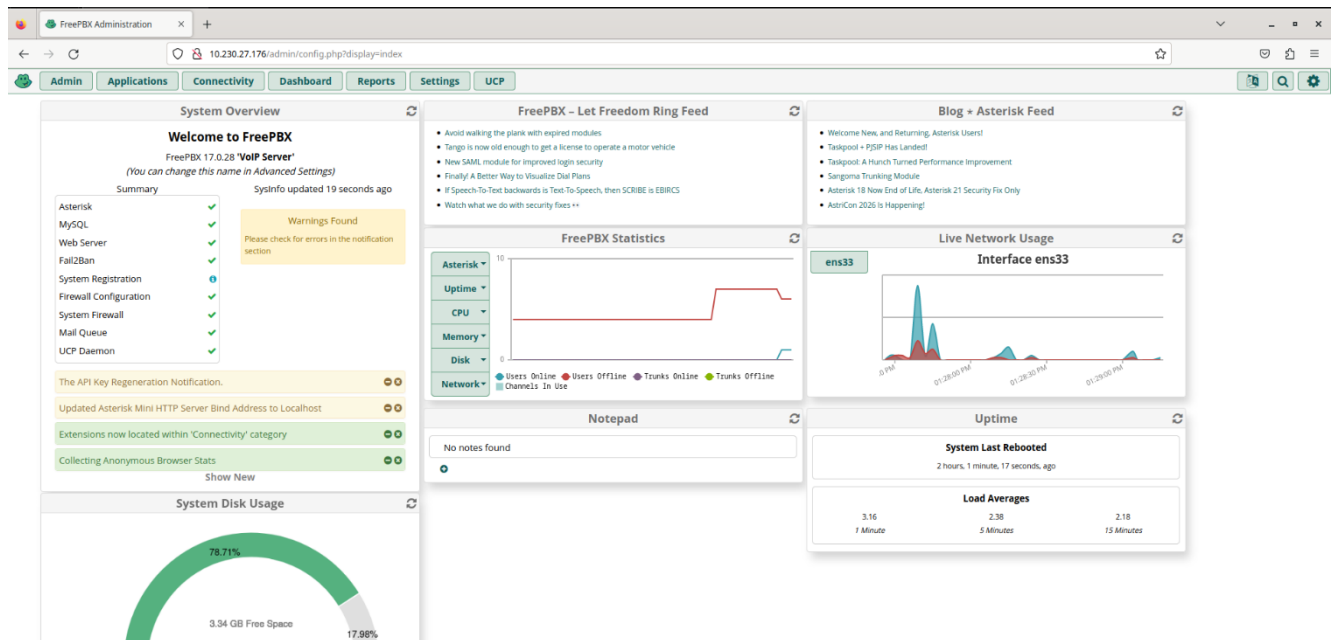


Figure 1: Top-level VoIP network architecture for the Sirs Systems PBX implementation.

2.3 PBX Server Environment

Component	Configuration
Hypervisor	VMware Workstation 17 Pro
Operating System	Debian 12
PBX Platform	FreePBX
Server IP Address	10.230.27.176



3. Dial Plan Design

3.1 Extension Plan

Extension Number	Purpose
2000	Reception
2100	Administration
2200	New Business Enquiries
2300	Business Support
6000	Staff Ring Group
6100	Support Ring Group
7000	Main IVR
8000	Conference Room
*97	Voicemail

3.2 Dial Plan Justification

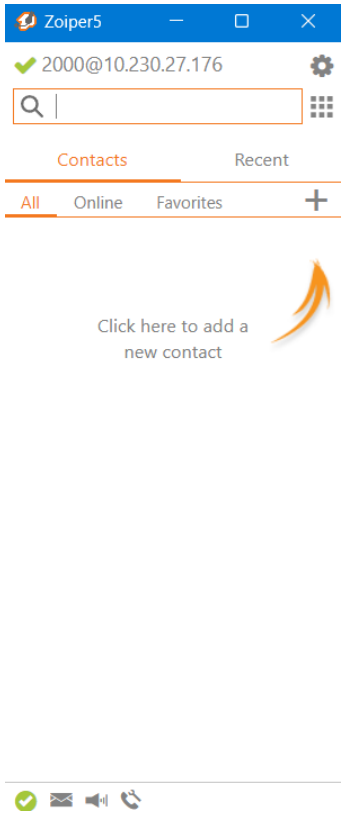
The dial plan was built utilising a systematic numbering method to increase scalability, usability and organisational clarity. The 2000 range was designated for individual SIP extensions, whereas the 6000 range was allocated for Ring Groups. The 7000 range was designated to IVR services and the 8000 range was reserved for conference facilities.

This framework facilitates future growth while retaining logical separation between consumers, services and organisational units.

4. Level 1 System Implementation

4.1 SIP Extension Configuration

Four SIP extensions were configured within the FreePBX environment to simulate geographically spread consumers.

Extension	User Role	Device
2000	Reception 	Zoiper 5 (Windows)
2100	Administration/Admin	Linphone (Windows)

A

General

Details

Edit your account information.

A

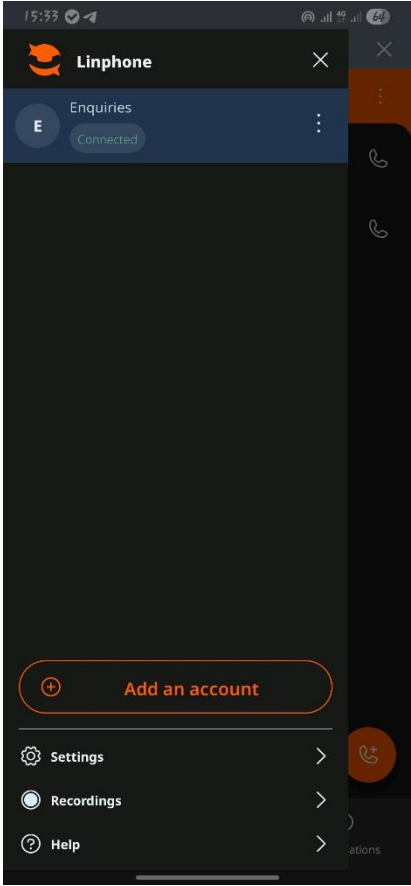
 Add an image

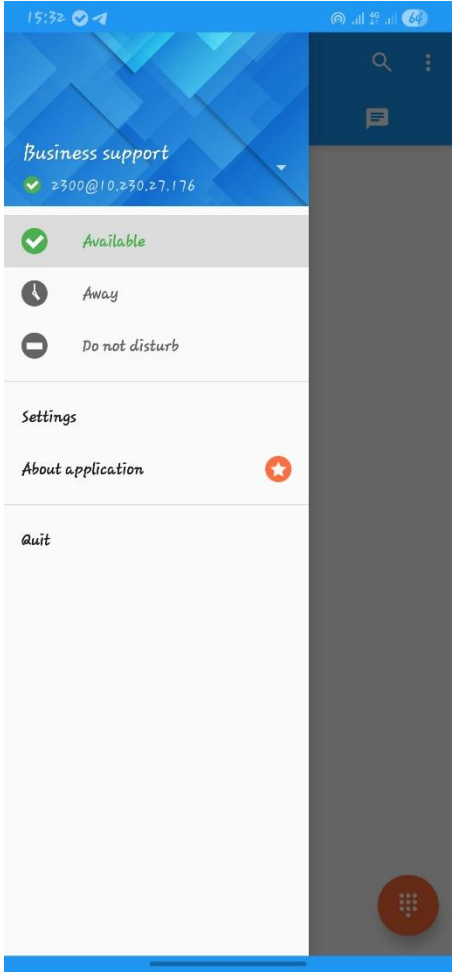
SIP address : sip:2100@10.230.27.176 

Display name
The name displayed to your contacts during exchanges.

Admin

International code*

2200	Enquiries 	Linphone (Android)
------	--	--------------------

2300	Business Support 	Sipnetic (Android)
-------------	---	---------------------------

Extension: 2000

General Voicemail Find Me/Follow Me Advanced Pin Sets Other

Edit Extension

This device uses **PJSIP** technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)

Display Name ⓘ	Reception
Outbound CID ⓘ	
Emergency CID ⓘ	
Secret ⓘ	pass2000

Language

Language Code ⓘ	Default
-----------------	---------

User Manager Settings

Linked to User 2000

Select User Directory: ⓘ	PBX Internal Directory
Link to a Different Default User: ⓘ	2000 (Linked)
Username ⓘ	☐ Use Custom Username
Password For New User ⓘ	
Groups ⓘ	All Users X

> Submit Reset De

Extension: 2100

General Voicemail Find Me/Follow Me Advanced Pin Sets Other

Edit Extension

This device uses **PJSIP** technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)

Display Name ⓘ

Administration

Outbound CID ⓘ

Emergency CID ⓘ

Secret ⓘ

pass2100

Language

Language Code ⓘ

Default

User Manager Settings

Linked to User 2100

Select User Directory: ⓘ

PBX Internal Directory

Link to a Different Default User: ⓘ

2100 (Linked)

Username ⓘ

☐ Use Custom Username

Password For New User ⓘ

g7*4803464.2100@2100-4536ad80775a-118a118a8080

Groups ⓘ

All Users X

> Submit Reset D

Extension: 2200

General Voicemail Find Me/Follow Me Advanced Pin Sets Other

Edit Extension

This device uses **PJSIP** technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)

Display Name ⓘ

Enquiries

Outbound CID ⓘ

Emergency CID ⓘ

Secret ⓘ

pass2200

Language

Language Code ⓘ

Default

User Manager Settings

Linked to User 2200

Select User Directory: ⓘ

PBX Internal Directory

Link to a Different Default User: ⓘ

2200 (Linked)

Username ⓘ

☐ Use Custom Username

Password For New User ⓘ

7a2200-4ad81f8080-8a-118a118a8080

Groups ⓘ

All Users X

> Submit Reset D

Extension: 2300

General Voicemail Find Me/Follow Me Advanced Pin Sets Other

— Edit Extension

This device uses **PJSIP** technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)

Display Name [?](#) BusinessSupport

Outbound CID [?](#)

Emergency CID [?](#)

Secret [?](#) [masked]

— Language

Language Code [?](#) Default

— User Manager Settings

Linked to User 2300

Select User Directory: [?](#) PBX Internal Directory

Link to a Different Default User: [?](#) 2300 (Linked)

Username [?](#) ☐ Use Custom Username

Password For New User [?](#) [masked]

Groups [?](#) All Users X

Submit Reset

Figure 2: SIP extension configuration within the FreePBX PBX environment

4.2 SIP Registration Testing

All configured SIP extensions were successfully registered with the FreePBX server. Internal VoIP connectivity between desktop and mobile SIP clients was validated satisfactorily.

10.230.27.176/admin/config.php?display=extensions#

Admin Applications Connectivity Dashboard Reports Settings UCP

All Extensions SIP (chan_pjsip) Extensions Virtual Extensions DAHDI Extensions IAX2 Extensions Custom Extensions

+ Add Extension Quick Create Extension Delete Search

Extension	Name	CW	DND	FM/FM	CF	CFB	CFU	Type	Actions
2000	Reception	☒	☐	☐	☐	☐	☐	pjsip	Edit Delete
2100	Administration	☒	☐	☐	☐	☐	☐	pjsip	Edit Delete
2200	Enquiries	☒	☐	☐	☐	☐	☐	pjsip	Edit Delete
2300	BusinessSupport	☒	☐	☐	☐	☐	☐	pjsip	Edit Delete

Figure 3: Successful SIP extension registration within the PBX system.

4.3 Voicemail Configuration

Voicemail functionality was enabled for all configured extensions to facilitate business continuity and distant accessibility.

Extension: 2000

General	Voicemail	Find Me/Follow Me	Advanced	Pin Sets	Other
---------	-----------	-------------------	----------	----------	-------

Voicemail

Enabled

YesNo

Voicemail Password

Set this password to same as extension number to force the user to setup their mailbox on first access.

Require From Same Extension

YesNo

Disable (*) in Voicemail Menu

YesNo

Email Address

reception@sirs.local

Pager Email Address

Email Attachment

YesNo

Play CID

YesNo

Play Envelope

YesNo

Delete Voicemail

YesNo

VM Options

VM Context

default

VmX Locator™

Enabled

YesNo

Use When:

UnavailableBusy

>

Submit

Reset

D

Extension: 2100

General	Voicemail	Find Me/Follow Me	Advanced	Pin Sets	Other
---------	-----------	-------------------	----------	----------	-------

Voicemail

Enabled

YesNo

Voicemail Password

Set this password to same as extension number to force the user to setup their mailbox on first access.

Require From Same Extension

YesNo

Disable (*) in Voicemail Menu

YesNo

Email Address

admin@sirs.local

Pager Email Address

Email Attachment

YesNo

Play CID

YesNo

Play Envelope

YesNo

Delete Voicemail

YesNo

VM Options

VM Context

default

VmX Locator™

Enabled

YesNo

>

Submit

Reset

De

4.3.1 Voicemail Access Testing

The voicemail retrieval feature was successfully tested using SIP softphones. The voicemail system was accessed internally by calling the FreePBX voicemail feature code.

Feature Code	Purpose
*97	Direct voicemail access

Testing was accomplished using the Sipnetic Android SIP client to validate effective voicemail authentication and message retrieval.

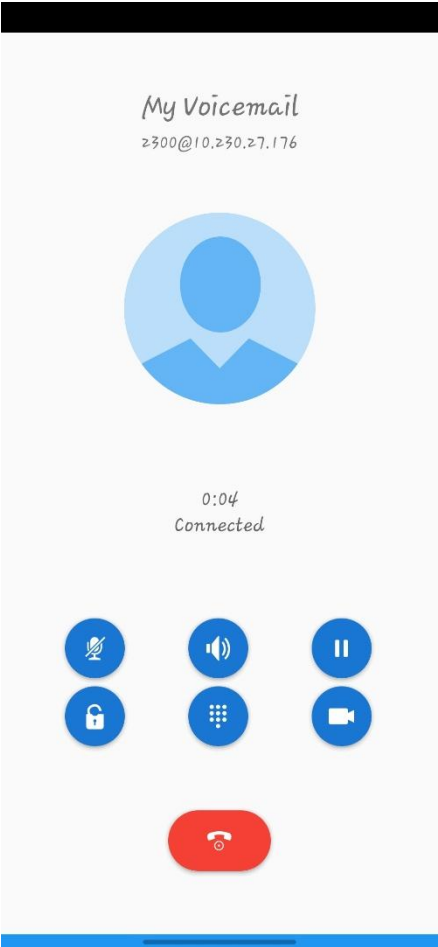


Figure 4: Successful voicemail retrieval with the *97 voicemail feature code.

The voicemail system allowed users to access stored messages remotely, boosting communication dependability and ensuring missed calls could be examined efficiently inside the VoIP context.

4.4 IVR / Auto-Attendant Configuration

An IVR system was setup to provide automatic call routing features for incoming callers.

The IVR menu included:

Menu Option	Destination
Press 1	Reception
Press 2	Administration
Press 3	Enquiries
Press 4	Business Support
Press 5	Conference Room

Edit Recording

Name ⓘ

SirsSystems_IVR

Description ⓘ

Main IVR greeting for Sirs Systems

TTS A.I Engine ⓘ

None

File List for English ⓘ

English

▶ custom/IVR

You can click any file above to replace it with a recording option below. Clicking a file will turn it green putting it into replace mode

Upload Recording ⓘ

Browse

Drop Multiple Files or Archives Here

Record In Browser ⓘ

▶ ● 00:00

00:00

Hit the red record button to start recording from your browser

Record Over Extension ⓘ

Enter Extension...

Call

Add System Recording ⓘ

Select a system recording

Link to Feature Code ⓘ

Yes No

Optional Feature Code *291

Feature Code Password ⓘ

Convert To ⓘ

alaw g722 g729 gsm sln

Submit Reset Delete

Admin Applications Connectivity Dashboard Reports Settings UCP

+ Add Recording

Search

⌵

Display Name	Description	Supported Languages	Actions
SirsSystems_IVR	Main IVR greeting for Sirs Systems	English	 

Showing 1 to 1 of 1 rows

Admin
Applications
Connectivity
Dashboard
Reports
Settings
UCP

Add IVR

IVR General Options

IVR Name

MainIVR

IVR Description

Sirs Systems Main Menu

IVR DTMF Options

Announcement

SirsSystems IVR

Enable Direct Dial

Enabled

Ignore Trailing # Key

Yes No

Force Strict Dial Timeout

Yes No No - Legacy

Timeout

10

Alert Info

None

Ringer Volume Override

None

Invalid Retries

3

Invalid Retry Recording

Default

Append Announcement to Invalid

Yes No

Return on Invalid

Yes No

Invalid Recording

Default

Invalid Destination

None

Timeout Retries

3

Submit
Duplicate
Reset

Append Announcement on Timeout

Yes No

Return on Timeout

Yes No

Timeout Recording

Default

Timeout Destination

None

Return to IVR after VM

Yes No

IVR Entries

Digits	Destination	Return	Delete
1	Extensions 2000 Reception	Yes No	
2	Extensions 2100 Administration	Yes No	
3	Extensions 2200 Enquiries	Yes No	
4	Extensions 2300 BusinessSupport	Yes No	
5	Conferences 8000 SirsMeetingRoom		

Add Another Entry

Submit
Duplicate

Figure 5: IVR auto-attendant configuration within FreePBX.

Create Internal IVR Extension (7000)

Admin
Applications
Connectivity
Dashboard
Reports
Settings
UCP

Misc Application

Enable

Yes No

Description

MainIVR

Feature Code

7000

Admin
Applications
Connectivity
Dashboard
Reports
Settings
UCP

Misc Application

Add Misc Application

Search

Description	Extension	Actions
MainIVR	7000	

Showing 1 to 1 of 1 rows

4.5 IVR Testing

The IVR system was tested successfully utilising internal SIP extensions. Call routing functionality performed correctly for various extension destinations.

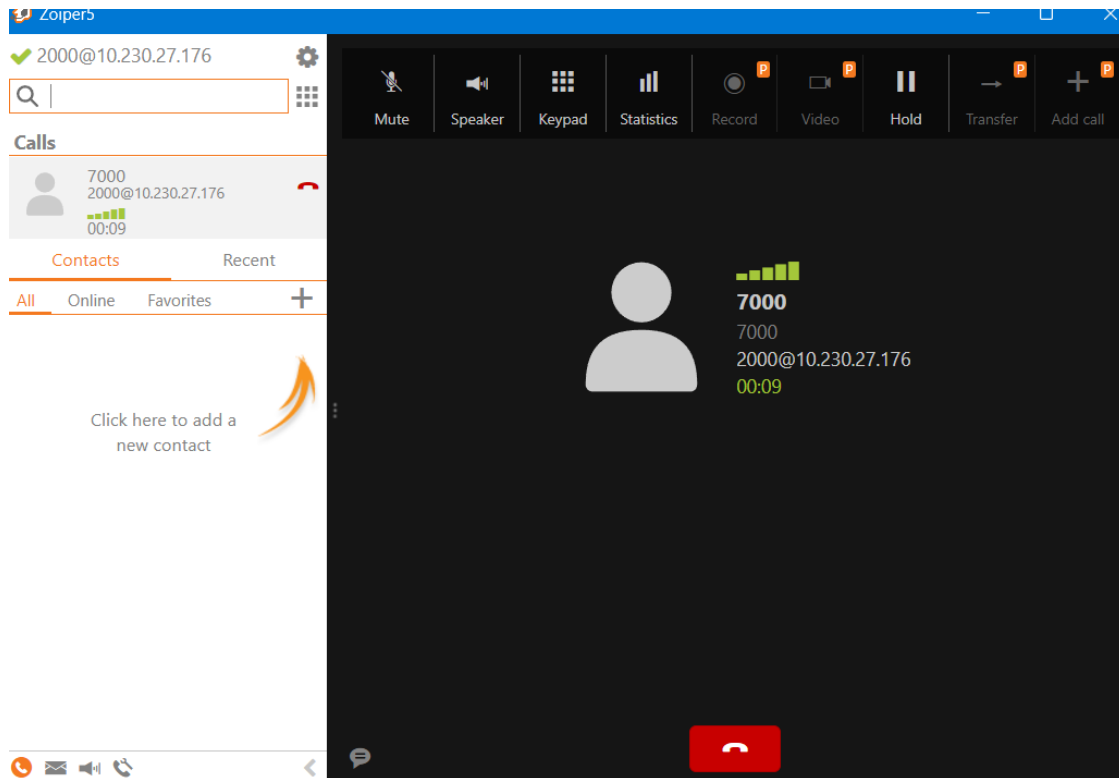


Figure 6: Successful IVR testing and internal call routing demonstration.

5. Level 2 System Implementation

5.1 Ring Group Configuration

Ring Groups were established to support geographically scattered workers functioning remotely across various sites.

❖ Staff Ring Group

Ring Group	Extensions
6000	2000, 2200

❖ Support Ring Group

Ring Group	Extensions
6100	2100, 2300

The ringall technique was adopted to facilitate simultaneous alerting across different SIP endpoints.

Ring Groups: Edit 6000

Group Description ⓘ

StaffRingGroup

14/35

Extension List ⓘ

2000

2200

User Quick Select ▾

Ring Strategy ⓘ

ringall

▾

Ring Time (max 300 sec) ⓘ

20

⌵

Announcement ⓘ

None

▾

Play Music On Hold ⓘ

Ring

▾

CID Name Prefix ⓘ

Alert Info ⓘ

None

⌵

Ringer Volume Override ⓘ

None

▾

Send Progress ⓘ

Yes

No

Mark Answered Elsewhere ⓘ

Always

Yes

No

Ignore CF Settings ⓘ

Yes

No

Skip Busy Agent ⓘ

Yes

No

Enable Call Pickup ⓘ

Yes

No

Confirm Calls ⓘ

Yes

No

Remote Announce ⓘ

Default

»

Submit

Reset

Del

Admin

Applications

Connectivity

Dashboard

Reports

Settings

UCP

Ring Groups

+ Add Ring Group

Search

Ring Group	Description	Actions
6000	StaffRingGroup	
6100	Support Ring Group	

Showing 1 to 2 of 2 rows

Figure 7: Ring Group configuration supporting simultaneous SIP endpoint ringing.

5.2 Ring Group Testing

Testing demonstrated that both desktop and mobile SIP clients rang simultaneously when Ring Group extensions were dialled.

The Administration extension successfully launched calls to both the Reception and Enquiries extensions using the set Ring Group (6000) functionality.

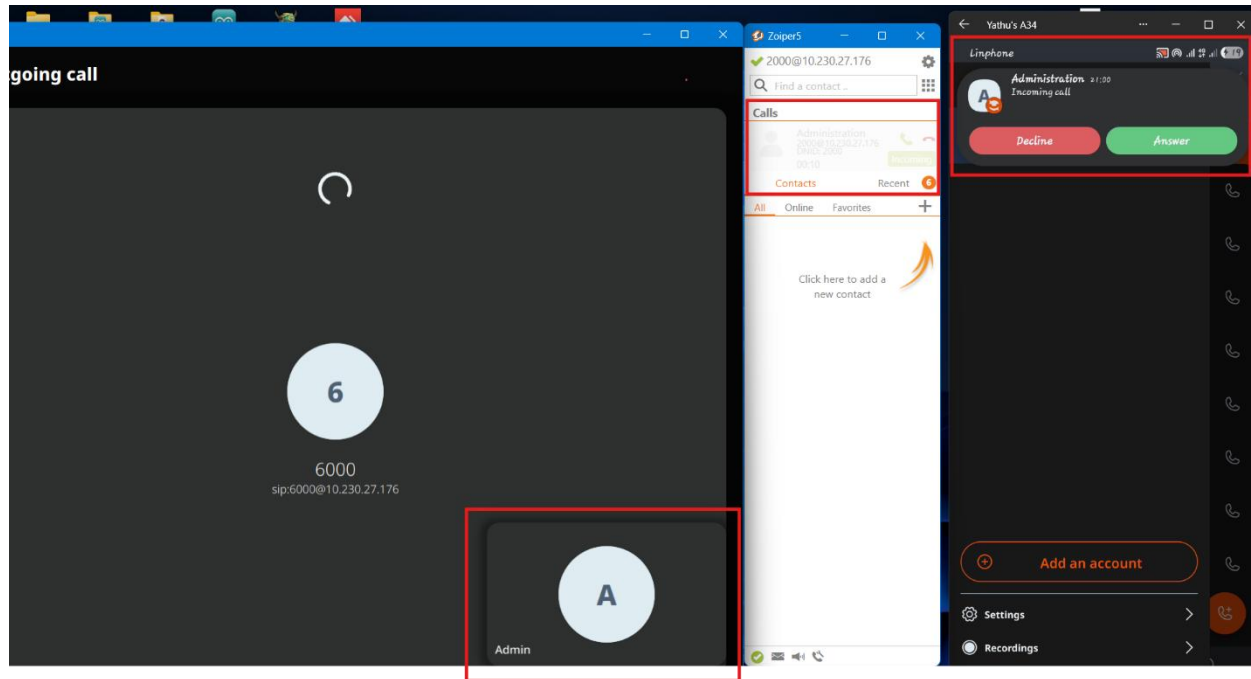


Figure 8: Successful Ring Group operation across several remote SIP devices.

5.3 Call Blocking / Blacklisting

Call blocking feature was built using the FreePBX blacklist module to prevent unwanted or malicious calls from accessing internal SIP extensions.

A test extension was added to the blacklist and call rejection behaviour was validated correctly.

For testing purpose, I block 2200 (Enquires) extension

Blacklist Module					
Admin Applications Connectivity Dashboard Reports Settings UCP					
Blacklist Import/Export Settings					
Delete Selected + Blacklist Number					
☐ Number Description Last Call Incoming Calls Actions ☐ 2200 BlockedExtension Tue, May 19, 2026 4:19 PM - 3 minutes ago 14 edit delete					
Showing 1 to 1 of 1 rows					

Figure 9: FreePBX blacklist setting used to block unwanted SIP calls.

6. Level 3 System Implementation

6.1 Audio Conferencing

Audio conferencing capabilities was created using the FreePBX Conferences module.

Conference Room:

Conference Number	Purpose
8000	SirsMeetingRoom

The conference function enabled several geographically scattered users to converse simultaneously through the PBX environment.

Conferences: Add

Conference Number ⓘ

8000

Conference Name ⓘ

SirsMeetingRoom

15/50

User PIN ⓘ

Admin PIN ⓘ

Language ⓘ

Inherit

▼

Join Message ⓘ

None

▼

Leader Wait ⓘ

Yes

No

Leader Leave ⓘ

Yes

No

Talker Optimization ⓘ

Yes

No

Talker Detection ⓘ

Yes

No

Quiet Mode ⓘ

Yes

No

User Count ⓘ

Yes

No

User join/leave ⓘ

Yes

No

Music on Hold ⓘ

Yes

No

Music on Hold Class ⓘ

Inherit

▼

Allow Menu ⓘ

Yes

No

Record Conference ⓘ

Yes

No

Maximum Participants ⓘ

0

⊞

Mute on Join ⓘ

Yes

No

Member Timeout ⓘ

21600

⊞

🌱 Admin Applications Connectivity Dashboard Reports Settings UCP

🔍 🔍 ⚙️

Conferences

+ Add ⓘ

Search

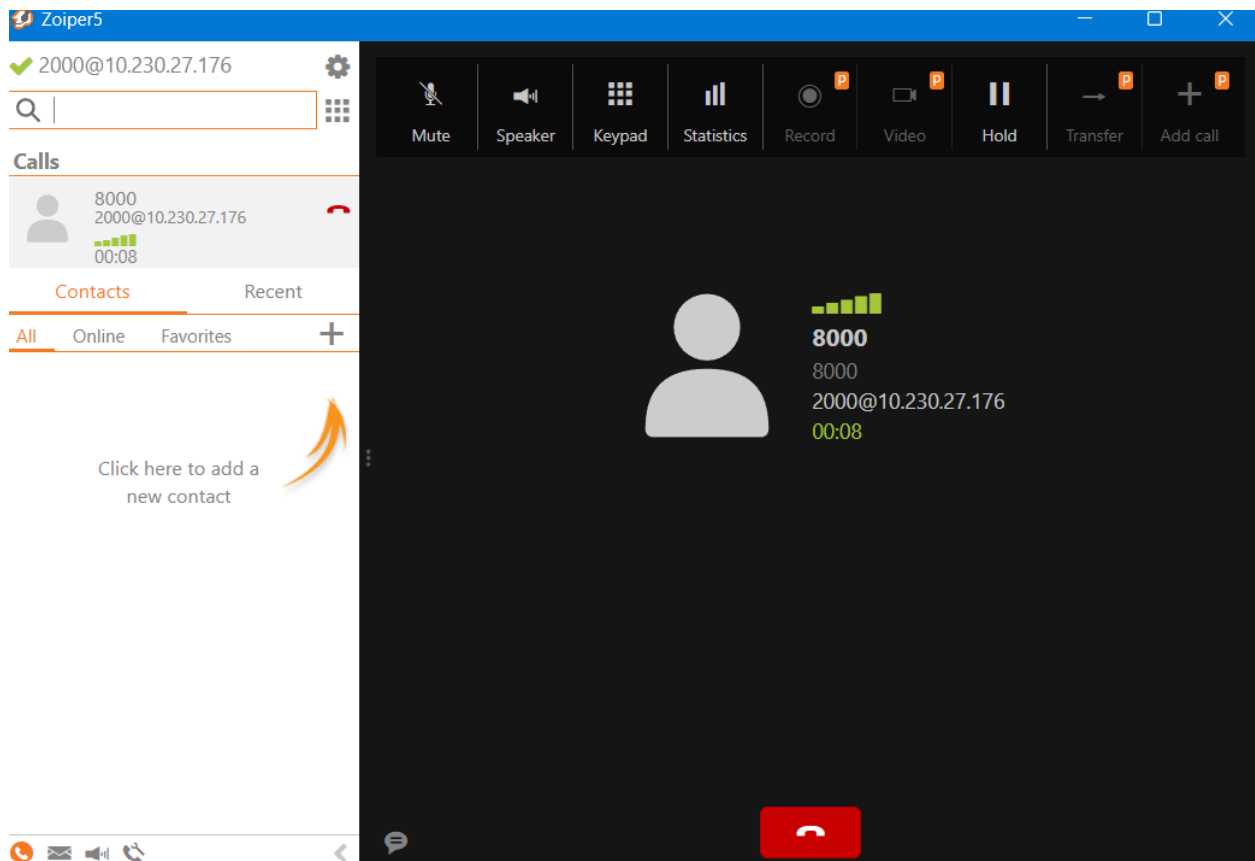
Conference	Description	Actions
8000	SirsMeetingRoom	✎️ 🗑️

Showing 1 to 1 of 1 rows

Figure 10: Conference room setting within the FreePBX PBX system.

6.2 Conference Testing

Conference functionality was successfully tested utilising desktop and mobile SIP softphones.



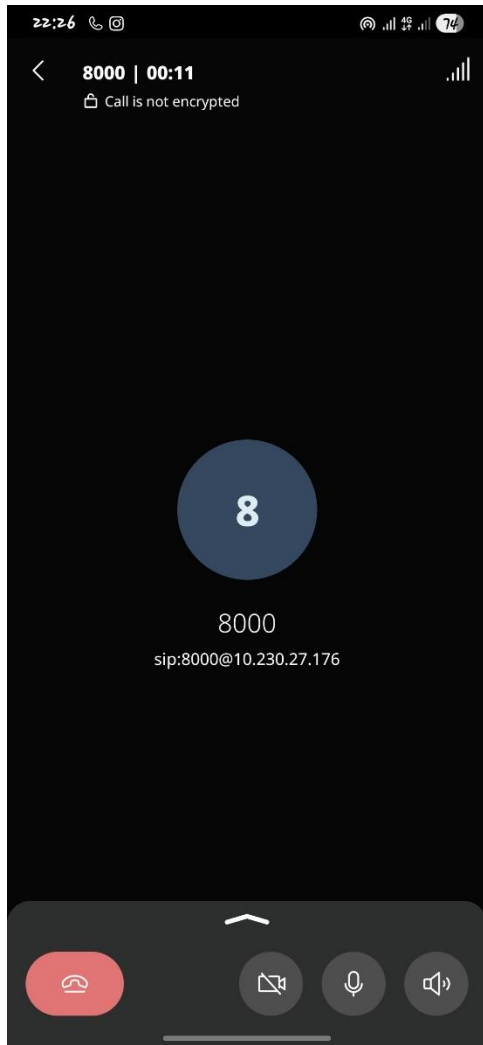


Figure 11: Active VoIP conference session between numerous SIP endpoints

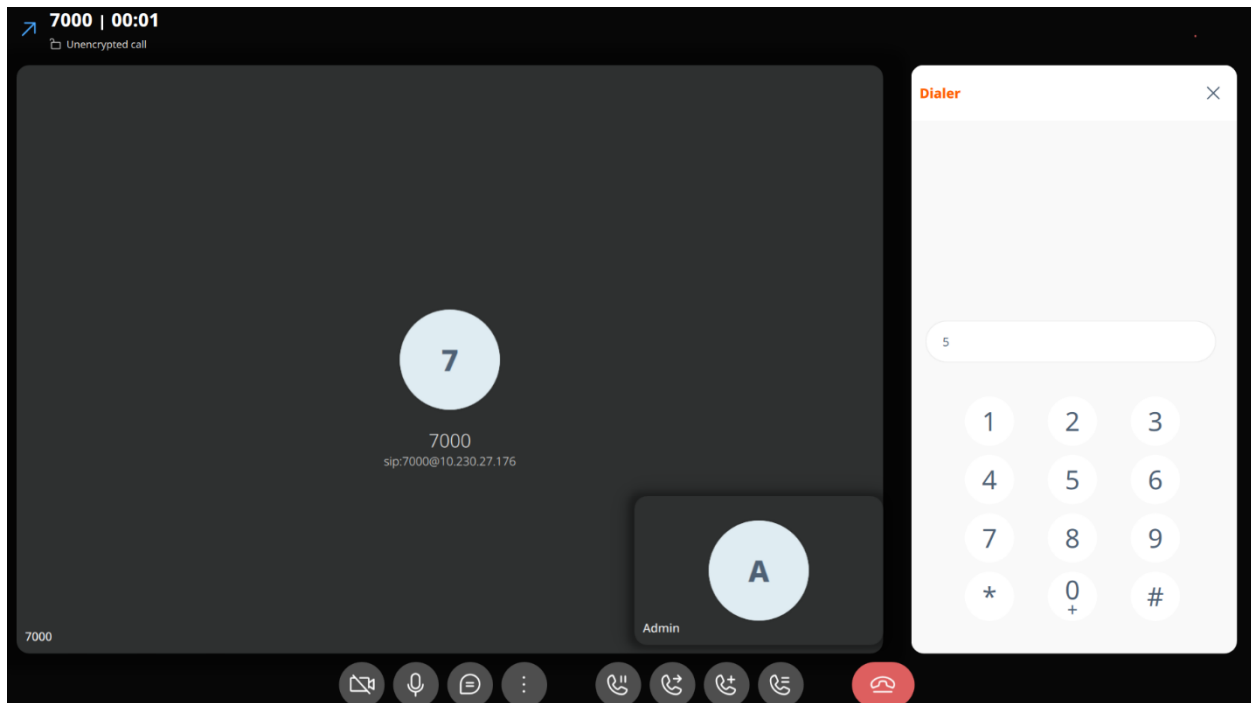


Figure 12: IVR menu integrated with the VoIP conference room to support remote collaborative communication.

6.3 VoIP Traffic Capture and Eavesdropping Demonstration

Wireshark packet analysis proved that normal SIP signalling communication can be viewed across the network when transmitted without encryption.

No.	Time	Source	Destination	Protocol	Length	Info
116	6.699881	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
117	6.699890	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
125	7.201213	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
126	7.201223	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
173	8.201417	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
174	8.201427	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
194	10.201795	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
195	10.201805	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
241	14.202985	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
242	14.202995	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
298	17.114230	10.230.27.176	10.182.75.171	SIP	515	Request: OPTIONS sip:2300@10.182.75.171:46294;x-reg=2C32736E208E4566
299	17.114242	10.230.27.176	10.182.75.171	SIP	515	Request: OPTIONS sip:2300@10.182.75.171:46294;x-reg=2C32736E208E4566
300	17.119962	10.182.75.171	10.230.27.176	SIP	618	Status: 200 OK (OPTIONS)
306	18.205948	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
307	18.205964	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
315	21.790896	10.230.27.176	10.230.27.139	SIP	523	Request: OPTIONS sip:2000@10.230.27.139:64245;rinstance=92d5882b9d75cbfa
316	21.790924	10.230.27.176	10.230.27.139	SIP	523	Request: OPTIONS sip:2000@10.230.27.139:64245;rinstance=92d5882b9d75cbfa
320	21.804951	10.230.27.139	10.230.27.176	SIP	747	Status: 200 OK (OPTIONS)
321	21.804960	10.230.27.139	10.230.27.176	SIP	747	Status: 200 OK (OPTIONS)
324	22.206731	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
325	22.206742	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
417	25.929265	10.230.27.176	10.230.27.139	SIP	468	Request: OPTIONS sip:2100@10.230.27.139:5060
418	25.929279	10.230.27.176	10.230.27.139	SIP	468	Request: OPTIONS sip:2100@10.230.27.139:5060
419	25.947375	10.230.27.139	10.230.27.176	SIP	337	Status: 200 OK (OPTIONS)
420	25.947384	10.230.27.139	10.230.27.176	SIP	337	Status: 200 OK (OPTIONS)
421	26.206544	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
422	26.206554	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
466	30.206835	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
467	30.206846	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572
477	33.044318	10.230.27.139	10.230.27.176	SIP/SDP	1066	Request: INVITE sip:2200@10.230.27.176;transport=UDP
478	33.044332	10.230.27.139	10.230.27.176	SIP/SDP	1066	Request: INVITE sip:2200@10.230.27.176;transport=UDP
479	33.045424	10.230.27.176	10.230.27.139	SIP	548	Status: 401 Unauthorized
480	33.045437	10.230.27.176	10.230.27.139	SIP	548	Status: 401 Unauthorized
481	33.046076	10.230.27.139	10.230.27.176	SIP	395	Request: ACK sip:2200@10.230.27.176;transport=UDP
482	33.046136	10.230.27.139	10.230.27.176	SIP	395	Request: ACK sip:2200@10.230.27.176;transport=UDP
483	33.163219	10.230.27.139	10.230.27.176	SIP/SDP	1364	Request: INVITE sip:2200@10.230.27.176;transport=UDP
484	33.163231	10.230.27.139	10.230.27.176	SIP/SDP	1364	Request: INVITE sip:2200@10.230.27.176;transport=UDP
485	33.165350	10.230.27.176	10.230.27.139	SIP	356	Status: 100 Trying

During testing, SIP INVITE and REGISTER messages exposed endpoint addressing information, extension numbers and session negotiation details.

sip.Method == "INVITE"						
No.	Time	Source	Destination	Protocol	Length	Info
3979	91.019397	10.230.27.139	10.230.27.176	SIP/SDP	1066	Request: INVITE sip:2300@10.230.27.176;transport=UDP
3980	91.019415	10.230.27.139	10.230.27.176	SIP/SDP	1066	Request: INVITE sip:2300@10.230.27.176;transport=UDP
3993	91.136462	10.230.27.139	10.230.27.176	SIP/SDP	1364	Request: INVITE sip:2300@10.230.27.176;transport=UDP
3994	91.136473	10.230.27.139	10.230.27.176	SIP/SDP	1364	Request: INVITE sip:2300@10.230.27.176;transport=UDP
4006	91.516027	10.230.27.176	10.180.124.235	SIP/SDP	1174	Request: INVITE sip:2300@10.180.124.235;41843;x-reg=216E2C42AA2BC48D
4007	91.516041	10.230.27.176	10.180.124.235	SIP/SDP	1174	Request: INVITE sip:2300@10.180.124.235;41843;x-reg=216E2C42AA2BC48D
Frame 3979: 1066 bytes on wire (8528 bits), 1066 bytes captured (8528 bits) on interface \Device\NPF{...} Ethernet II, Src: ChongqingFug_e4:46:75 (a4:97:b1:e4:46:75), Dst: VMware_a0:7d:5e (00:0c:29:00:00:00) Internet Protocol Version 4, Src: 10.230.27.139, Dst: 10.230.27.176 User Datagram Protocol, Src Port: 64351, Dst Port: 5060 Session Initiation Protocol (INVITE)						
Request-Line: INVITE sip:2300@10.230.27.176;transport=UDP SIP/2.0						
Method: INVITE						
Request-URI: sip:2300@10.230.27.176;transport=UDP [Resent Packet: False]						
Message Header						
Via: SIP/2.0/UDP 10.230.27.139:64351;branch=z9hG4bK-524287-1---289ada7e29afeafa;rport=...						
Max-Forwards: 70						
Contact: <sip:2000@10.230.27.139:64351;transport=UDP>						
To: <sip:2300@10.230.27.176>						
SIP to address: sip:2300@10.230.27.176						
SIP to address User Part: 2300						
SIP to address Host Part: 10.230.27.176						
From: <sip:2000@10.230.27.176;transport=UDP>;tag=5c707731						
SIP from address: sip:2000@10.230.27.176;transport=UDP						
SIP from tag: 5c707731						
Call-ID: ansTKJQUU0170Mj0653B8A..						
[Generated Call-ID: ansTKJQUU0170Mj0653B8A..]						
Cseq: 1 INVITE						
Allow: INVITE, ACK, CANCEL, BYE, NOTIFY, REFER, MESSAGE, OPTIONS, INFO, SUBSCRIBE						
Content-Type: application/sdp						
Supported: replaces, noanswersub, extended-refer, timer, sec-agree, outbound, path, X-User-Agent: 7.5.6.13 v2.10.20.14						
Allow-Events: presence, kpml, talk, as-feature-event						
Content-Length: 343						
Message Body						

Figure 13: SIP signalling traffic obtained using Wireshark.

6.4 VoIP Security Configuration

VoIP security measures were studied within the FreePBX PBX environment through the implementation of TLS-based SIP transport and SRTP media encryption settings. Although full end-to-end encrypted communication was not regularly achieved due to interoperability issues between SIP softphone applications and self-signed certificates, the implementation effectively demonstrated the principles of secure VoIP communication and SIP transport security.



Figure 14: During testing, the softphone applications indicated that active calls remained unencrypted, underscoring the significance of secure VoIP protocols such as TLS and SRTP for protecting voice communication from interception and eavesdropping assaults.

6.5 Firewall Configuration

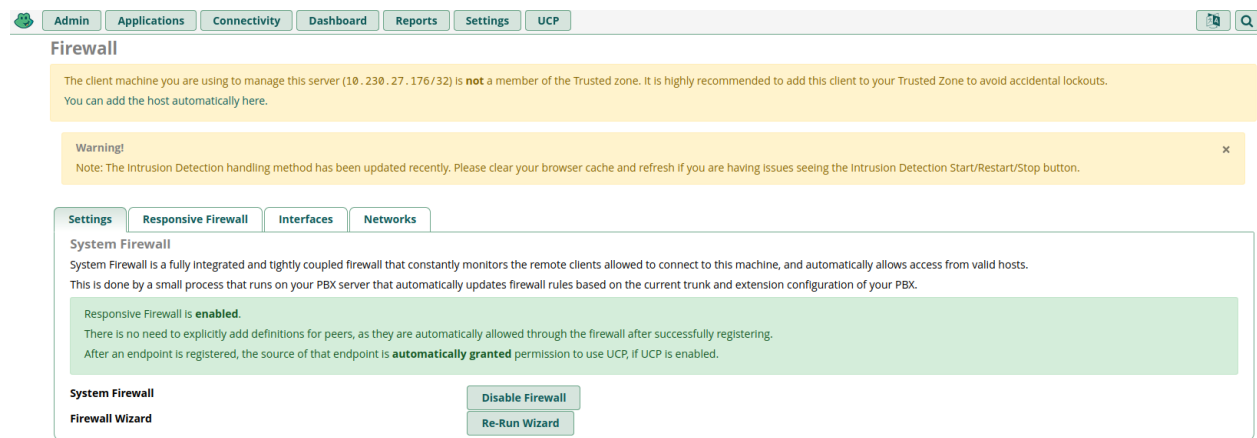


Figure 15: FreePBX firewall setup used to increase PBX security and network access management.

The built-in FreePBX Firewall module was enabled to increase the overall security of the PBX environment. The firewall provides extra protection by preventing unauthorised access attempts to SIP services and PBX administration interfaces. Trusted local network zones were permitted but external access remained regulated by the FreePBX security settings. This additional security layer helps lower the risk of hostile VoIP attacks and unauthorised system access.

7. Additional Features

7.1 Music on Hold

An additional Music on Hold (MOH) capability was integrated into the FreePBX PBX system to enhance the caller experience during wait and transfer procedures.

The default MOH class supplied by FreePBX was employed and successfully tested in Ring Group and call handling scenarios.

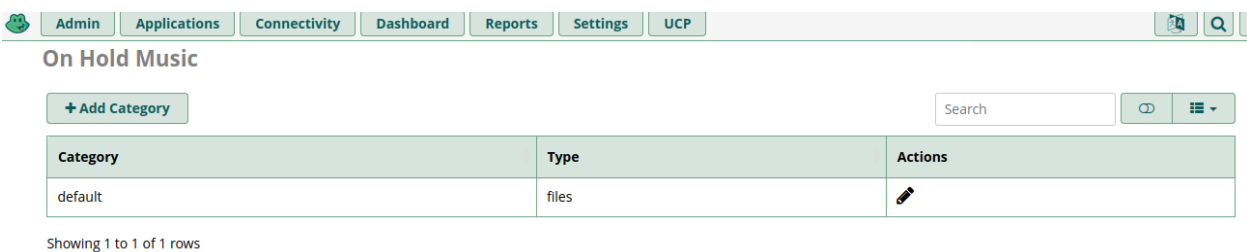


Figure 16: Default Music on Hold settings are accessible within the FreePBX PBX system.

8. Testing and Results

Feature	Tested	Result
SIP Registration	Yes	Successful
Internal Calling	Yes	Successful
Voicemail	Yes	Successful
IVR	Yes	Successful
Ring Groups	Yes	Successful
Conferencing	Yes	Successful
Blacklisting	Yes	Successful
Wireshark Packet Capture	Yes	Successful
Music on Hold	Yes	Successful

9. Conclusion

The practical implementation successfully demonstrated the deployment and configuration of a complete VoIP PBX environment utilising FreePBX and Asterisk technologies. The completed system featured basic business communication features including SIP extensions, voicemail, IVR call routing, Ring Groups, conferencing facilities, call blocking and VoIP security configuration.

The project proved how VoIP technology may serve geographically spread businesses through flexible and cost-effective communication solutions running across desktop and mobile platforms. In addition, the research studied VoIP security topics including SIP traffic analysis, secure transport configuration and the relevance of encrypted communication protocols.

Overall, the entire solution fulfilled the objectives of the Sirs Systems scenario while exhibiting practical mastery of current VoIP deployment, configuration and security principles inside a virtualised network environment.